

2022-Ph-H1-Q23

Section: Electricity

Topic: Current, PD, Power, Resistance

A resistor of resistance $4\ \Omega$ carries a current of $3\ \text{A}$. What is the power dissipated?

Answer: B

(36 W)

Guidance for Students:

Use the formula:

$$P = I^2 R = 3^2 \times 4 = 9 \times 4 = 36\ \text{W}$$

Revision Tips:

- Three formulas: $P = IV$, $P = I^2 R$, $P = \frac{V^2}{R}$
- Pick the one that fits the values given
- Power is energy transferred per second